

# NIKIL KRISHNA *Undergraduate Student*

✉ 126010091@sastra.ac.in    📍 SASTRA Deemed University

🔗 nikilkrishna.netlify.app    🌐 nikil-krishna-0b2a5b1bb



## ABOUT ME

---

My academic and research interests lie at the intersection of biotechnology and computational biology. My experience encompasses molecular visualization, design, and simulation. I am currently learning about artificial intelligence and machine learning applications for in silico drug design and protein binder design. Outside of academia, I enjoy listening to music, reading, and engaging in various forms of craftwork.

## EXPERIENCE

---

### Capstone Project and Publication

04/2026

SASTRA Deemed University

- Structure-Based Drug Design of Novel Inhibitors Targeting the Thiamine Biosynthesis Pathway in Bacteria
- High throughput **Virtual Screening**
- **Lead Optimization** by R-Group substitution and SAR analysis
- Molecular Dynamics and Analysis
- Preprint available at

### Founder

01/2026

Phage

- **Molecular Dynamics** as a Service.
- Frontend **web design**
- Production Grade Molecular Dynamics **Automation**
- Scaling, Optimization and Maintenance
- **Agentic Engineering**

### Summer Internship

06/2025

IIT Dharwad

- Development of Early-Stage Computational **Drug Discovery Pipeline** Using **Python**
- Virtual Screening, Rescoring and Statistical Analysis **Automation**
- Virtual Environment and dependency management using **Conda**
- Parallel High Performance **Scientific Computing**
- Benchmark of three SOTA **machine learning** architectures for protein binder design

### Undergraduate Student Researcher

06/2024

SASTRA Deemed University

- Molecular Motors Laboratory
- Drug Design and Protein Engineering

## EDUCATION

---

### B.tech Biotechnology

2022

SASTRA Deemed University, Thanjavur

### HSC and SSLC

2020

ALPHA Wisdom Vidyashram

## SKILLS

---

### ML Protein Structure Prediction

- AlphaFold
- Boltz
- Diffusion Based Models

### Drug Design

- Virtual Screening
- Fragment Based Drug Design
- Genetic algorithm/Combinatorial Chemistry
- Lead Optimization
- ADMET analysis

### Molecular Docking and Dynamics

- Autodock
- DiffeDock
- Gromacs
- OpenMM

### Process Simulation

- Aspen Plus
- SuperPro
- Designer
- DWSIM

### Molecular Modelling

- ChimeraX
- Pymol
- VMD
- MOE
- Schrodinger Maestro

### Python

- Molecular Visualization
- Data Analysis
- Machine Learning
- Bioinformatics Pipelines
- Dependency management
- AI Automation

### Bioinformatics

- Linux
- BLAST, MSA
- Visual Studio Code
- Git version management
- Scientific Computing

## LANGUAGES

---

### English

Professional working proficiency

### Hindi

Limited working proficiency

### Tamil

Bilingual proficiency

### Kannada

Bilingual proficiency

### Telugu

Elementary proficiency

## CERTIFICATIONS

---

### NPTEL

Aspen Plus Simulation  
Software - A Basic Course  
For Beginners

- 90%

### Schrodinger Drug

#### Discovery Hackathon Winner

Insilco Structure-based Drug  
Design targeting  
*Mycobacterium Tuberculosis*  
using genetic algorithm and  
combinatorial chemistry.

### NPTEL

2024 Drug Delivery:  
Principles and Engineering

- 83%